eg

N = 4 → Rows

```
* * * *
* * * *
* * * *
* * * *
```

Columns

2

might not be same

for each row

↓

How many columns to print ith row?

What to print?

```
n = int (input())
i = 1
while i <= n:
    print ('#', end = ' ')
    print Ith row
    j = 1
    while j <= n:
        # print jth column.
        j = j + 1
    print()
    i = i + 1
```

empty string

If we want in same line

for going to next row.

If we write end = '1'

2

then

*1*1*1

Character Patterns

ABCD

ABCD

ABCD

ABCD

'A' + 1

'A' + j - 1

we store their ASCII value.

Kth alphabet

k = int (input())

ascii + (k-1)

ord ('A') + k - 1

A → 65

B → 66

chr (66) $\xrightarrow{o/p}$ B

chr gives character corresponding to ASCII and ord gives ASCII corresponding to character.

i = 0

- - - * → j = N - i - 1

- - * * → j = N-1, N

- * * * i = 1

3

0 0 0 1 0 0 Printing till i+1 at j = N-1

0 0 1 2 1 0 0

0 1 2 3 2 1 0

1 2 3 4 3 2 1

```
for i in range (N)
    a = 1
    for j in range (N*2-1)
        if j < N-i-1 or
        j > N-i-1
            print (space _
        else :
            print (a
            a +=1
            if a > i+1
                a = i+1
                a -= 1
```

My

4-0-1

3

Sol^n → Don't worry about spaces at last.

```
n = int(input ())
i = 1
while i <= n :
    spaces = 1
    while spaces <= n - i :
        print ("_" , end = " ")
    p = 1
    j = 1
    while j <= i
        print (p , end = " ")
        j = j+1
```

```
        p = p+1
p = i-1
while p>=1:
    print (p, end = " ")
    p = p-1
print ()
i = i+1.
```

$2N-i-1$

$8-0-1$

Q $\xrightarrow{I/P}$ 3

1 2 3 4

1 2 3

1 2

1

P

→ N=4

Rows → N

Columns → $N*2-1$

1 → 6

2 → 4

3 → 2

4 → 0

a = 1

7

j < 2*N-1

j = 0

0 1 2 3

j < 7

```
for i in range (N)
    spaces = 1
    for j in range (2*N-1)
        a = 1
        if spaces < 2N-i-1
            print (a)
            a+1
        else print (space)
```